Mend-Amar Badral

MendeBadra | Mend-Amar Badral | mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

Skills

Languages:	English (C1 IELTS 8.0)
Programming:	Python, Bash, Julia, C/C++, R
Tools:	Linux, Git, Docker, SSH, Tailscale, pandas, sklearn, tidyverse, PyTorch, Quarto, LaTeX
Technical Areas:	Applied mathematics, statistics, data analysis, technical documentation, interactive visualization, image processing, 3D computer vision

Education

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: **Evaluating land degradation using image processing methods**

Experience

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Programming & Data Analysis:** Developed a custom **Python application** for processing a total of 1800 multispectral drone imagery dataset capture collected from 5 field areas and calculating vegetation indices and K-means segmentation maps for the vegetation. Utilized multi-threading, made reduction in processing times up to a factor of 5.
- **Documentation & Teamwork:** Authored documentation, user guides, and operational procedures for research workflows, improving onboarding and collaboration within a student group.
- **System administration:** Maintained remote access to laboratory servers and computers using Tailscale and SSH, managed and distributed SSH keys, configured automated backups.

Projects

aiSim Data Integration for 3D Gaussian Splatting

- Developed a Python-based data integration pipeline to parse and convert over sensor measurements from aiMotive's aiSim simulation platform into the nerfstudio 3D scene representation format.

Exploratory Data Analysis – Bike Sharing Dataset

- Performed exploratory data analysis on bike sharing dataset. Published the results personal blog website using Quarto.

Deep Learning Foundations and Concepts Book Examples Reproduction

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

Awards

Stipendium Hungaricum Scholarship Sep, 2025